

Sejal Girish Hukare

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EDUCATION

Master of Science in Data Science, 3rd Semester <i>University of Colorado Boulder</i> Relevant Coursework: Machine Learning for Data Science, Data Mining, Statistical Methods & Analysis, Relational Database Design & SQL, Information Visualization, Cybersecurity in Data Science	Aug 2025 – Expected Graduation 2027
Bachelor of Technology in Artificial Intelligence and Data Science, GPA: 8.60/10.00 (~3.4/4.0) <i>Vishwakarma Institute of Technology, Pune, India</i> Relevant Coursework: Data Mining and Knowledge Discovery, Business Intelligence and Analytics, Artificial Intelligence, Machine Learning, Cloud Technologies	Aug 2020 – May 2024

TECHNICAL SKILLS

Programming Languages:	Python, SQL, Java, JavaScript
ML / Data Science:	Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Prophet, Random Forest, SVM, Logistic Regression, Naïve Bayes, K-Means, Association Rule Mining
Data Visualization / BI:	Tableau, Folium, Metabase, Grafana
Cloud & Tools:	AWS, Git, Docker, Postman
Databases:	MySQL, HeidiSQL, MongoDB

WORK EXPERIENCE

ERPNext Full Stack Developer Medsynaptic Pvt. Ltd. • Maintained the company's Frappe/ERPNext installation running on the local office network — configuration, custom workflows, and fixes as they came up, largely independently. • Designed and shipped a RESTful API layer, tested through Postman, connecting the front end to a HeidiSQL database — cutting manual data entry by 30%. • Diagnosed and resolved backend issues directly against the live database when workflows broke, keeping downtime minimal for daily operations. • Built Metabase and Grafana dashboards on top of that same data, giving the team visibility they didn't have before.	Jan 2025 – Jun 2025
Python Developer Intern Medsynaptic Pvt. Ltd. • Developed a HIS-PACS integration system to streamline hospital data transfer. • Optimized DICOM data processing, improving retrieval speed by 30%. • Implemented SQL query optimization, reducing report generation time by 40%.	Jun 2024 – Dec 2024

PROJECT EXPERIENCE

Walkability Across America — EPA National Walkability Index • Analyzed 220,000+ US census block groups using clustering, association rule mining, Naïve Bayes, SVM, logistic regression, and a 200-tree Random Forest to identify what separates walkable neighborhoods from car-dependent ones. • Found transit access to be the single strongest predictor across every method tested; Random Forest reached 86%+ accuracy. • Deployed an interactive Streamlit app for public exploration of the results.	2025 – Present
Sentiment Analysis of Twitter Data on the Russia-Ukraine Conflict • Evaluated the predictive accuracy of five machine learning models, including Random Forest and SVC, for misinformation and sentiment classification. • Built an ML pipeline using Logistic Regression, Naïve Bayes, and SVM; achieved 87% accuracy with TF-IDF, a 12% improvement over bag-of-words. • Analyzed 500,000 tweets to identify sentiment trends and misinformation spread.	Jan 2024 – Apr 2024
Crime Analysis and Crime Prediction of Toronto • Engineered a Random Forest and K-Means clustering model to forecast Toronto crime hotspots, achieving 85% accuracy. • Implemented Prophet forecasting, improving assault prediction accuracy by 15%. • Mapped crime heatmaps using Folium and Tableau for visualization.	Aug 2022 – Dec 2022
Audio Deepfake Detection Using AI • Converted audio spectrograms into training data for a 3-layer CNN model. • Achieved 78% accuracy using categorical cross-entropy loss and the Adam optimizer, a 20% improvement in fake-audio detection over baseline.	Jul 2023 – Nov 2023

PUBLICATIONS

Facial Emotion Based Song Recommendation System	<i>IEEE CISES, Apr 2023</i>
Personalized Travel Recommendation System: A Hybrid Model Based on Ratings and Image Analysis	<i>AIP Conference Proceedings, Jan 2023</i>
Survey on Banker's Algorithm to Remove Deadlock	<i>River Publishers, 2022</i>
Dijkstra's Shortest Path Visualization	<i>IRJET, Jan 2022</i>

CERTIFICATIONS

- Data Analyst (Internship and Training Program) — SHAPEAI
- Computer Vision — CHARUSAT